Unit 3- 5

Posets, Hasse Diagram and Lattices.

Generating functions, solution by method of generating function Recurrence Relation and generating Function : Introduction to Recurrence Relation, Linear Recurrence Relation with Constant Coefficient, Homogeneous solutions, Particular Solution Total solutions :.

GENERATING Function :-

Generating functions provide us an Alternative way to represent discrete numeric functions and solving recurrence relation. and it also provide us an alternative way to solve combinatorial problem.

Let, $(a_0, a_1, a_2, \ldots a_r \ldots) = a$ be a discrete numerical function, then an infinite series in terms of parameter Z, written as

$$A(z) = a_0 + a_1 z + a_2 z^2 + \ldots + a_r z^r + \ldots \quad -(1)$$

A(z) is said to be a generating function of the numeric funct^n a.

Sequence → $a_0, a_1, a_2 \ldots a_k \ldots = x$

$$G(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 + a_3 x^3 \ldots + a_k x^k + \ldots$$

Example :-

(1) $1, 1, 1, 1, \ldots = w$

$$G(w) = 1 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 \cdot 1 \cdot x^3 \ldots + 1 \cdot x^k + \ldots$$

$$= 1 + x + x^2 + \ldots x^k + \ldots$$

(2) $1, 2, 3, 4, 5, 6 \ldots = N$

$$G(N) = 1 + 2x + 3x^2 + 4x^3 + \ldots$$

③ $1, -1, 1, -1 \ldots\ldots \quad - k$

$G(k) = 1 - x + x^2 - x^3 + x^4 - \ldots\ldots$

④ $2, 2, 2, 2, \ldots\ldots$

⑤ $0, 1, 2, 3, \ldots\ldots$

⑥ $1, a, a^2, a^3 \ldots\ldots$

⑦ $1, -a, a^2, -a^3 \ldots\ldots$

※ Closed form of generating function :-

① $1, 1, 1, 1, \ldots\ldots$ OR $a_k = 1$ $\quad - x$

$$\left\{ GP(\text{geometric Progression}) = a + ar + ar^2 + ar^3 + \ldots\ldots = \frac{a}{1-r} \right\}$$

r = common ratio

$G(x) = 1 + x + x^2 + x^3 + \ldots\ldots$

$$\boxed{G(x) = \frac{1}{1-x}}$$

② $2, 2, \ldots\ldots$ OR $a_k = 2$

$G(x) = 2 + 2x + 2x^2 + 2x^3 + \ldots\ldots$

$G(x) = 2(1 + x + x^2 + x^3 + \ldots\ldots)$

$= 2 \cdot \left[\frac{1}{1-x}\right]$

$$\boxed{G(x) = \frac{2}{1-x}}$$

The generating functions for some simple function are tabulated in the following table :-

S-No.	a_k (the general term of numeric $funct^n$)	Generating function $G(x)$
1.	$a_k = 1$	Sq:- 1, 1, 1 ...... $G(x) = 1 + x + x^2 + \ldots + x^k + \ldots$ $\boxed{G(x) = \frac{1}{(1-x)}}$
2.	$a_k = (-1)^k$	Sq :- 1, -1, 1, -1, ... $G(x) = 1 - x + x^2 - x^3 + \ldots$ $\boxed{G(x) = \frac{1}{1+x}}$
3.	$a_k = c$	Sq :- $c + cx + cx^2 + cx^3 + \ldots$ $\boxed{G(x) = \frac{c}{1-x}}$
4.	$a_k = k+1$	Sq :- 1, 2, 3, 4 ...... $\boxed{G(x) = \frac{1}{(1-x)^2}}$
5.	$a_k = k$	Sq :- 0, 1, 2, 3 ...... $\boxed{G(x) = \frac{x}{(1-x)^2}}$
6.	$a_k = \alpha^k$ where, α is a constant	Sq :- $1 + \alpha x + \alpha^2 x^2 + \alpha^3 x^3 + \ldots$ $\boxed{G(x) = \frac{1}{1-\alpha x}}$
7.	$a_k = (-\alpha)^k$ where α is constant	$\boxed{G(x) = \frac{1}{1+\alpha \cdot x}}$
8.	$a_k = k^2$	$G(x) = x + 2^2 x^2 + 3^2 x^3 + 4^2 x^4 + \ldots k^2 x^k \ldots$ $\boxed{G(x) = \frac{x(x+1)}{(1-x)^3}}$

9.	$a_k = k \cdot a^k$	$\boxed{G(x) = \dfrac{a \cdot x}{(1-ax)^2}}$
10.	$a_k = k(k+1)$	$G(x) = 1\cdot 2x + 2\cdot 3x^2 + 3\cdot 4x^3 + \dots + k(k+1)x^k + \dots$ $G(x) = \dfrac{2x}{(1-x)^3}$

Some more Sequences :-

* $x = 0, 0, 1, 1, 1 \dots\dots$

$$G(x) = 0 + 0\cdot x + 1\cdot x^2 + 1\cdot x^3 + \dots\dots$$
$$= x^2 + x^3 + x^4 + \dots$$
$$= x^2 [1 + x + x^2 + \dots\dots]$$
$$\boxed{G(x) = \frac{x^2}{1-x}}$$

* $x = 0, 0, 0, 1, 1, 1, \dots\dots$

$$\boxed{G(x) = \frac{x^3}{1-x}}$$

Que 3:- Find the closed form of the generating function

* $a_n = 3$
* $a_k = k+3$
* $a_k = 3^k$
* $a_k = k(3+5k)$ {you have to solve it in detail}
* $a_k = k(k-1)$

Sol:- $a_n = 3$, Seq = 3, 3, 3, 3, 3 $\Rightarrow G(x) = 3x^0 + 3x^1 + 3x^2 + 3x^3 \dots$

$$G(x) = \frac{3}{1-x}$$

$a_k = k+3 \Rightarrow$

Seq $= \{0, 1, 2, 3, 4 \dots\} + 3, 3, 3, 3 \dots$

$G(x) = \{0 \cdot x^0 + x^1 + 2 \cdot x^2 + 3 \cdot x^3 \dots\} +$
$\{3x^0 + 3x^1 + 3x^2 + 3x^3\}$

$$G(x) = \frac{x}{(1-x)^2} + \frac{3}{1-x}$$

$G(x) = \{\cancel{1} + x + 2x^2 + 3x^3 + \dots\} +$
$3\{1 + x + x^2 + x^3 \dots\}$

$$G(x) = \frac{x}{(1-x)^2} + \frac{3}{(1-x)}$$

$a_k = 3^k$

$$G(x) = \frac{1}{1-3x}$$

$a_k = k(3+5k) = 3k + 5k^2$

$$G(x) = \frac{3x}{(1-x)^2} + \frac{5x(1+x)}{(1-x)^3}$$

$a_k = k(k-1) = k^2 - k$

$$G(x) = \frac{x(1+x)}{(1-x)^3} - \frac{x}{(1-x)^2}$$

Que : Find the closed form of the generating function

1) $0, 0, 0, 1, 1, 1 \text{ -----}$

2) $0, 0, 1, 2, 3 \text{ -----}$

3) $3, -3, 3, -3, \text{ -----}$

4) $3, 9, 27, 81 \text{ -----}$

5) $1, -2, 3, -4 \text{ -----}$

Sol :- 1) $x = 0, 0, 0, 1, 1, 1 \text{ -----}$

$$G(x) = 0 + 0x + 0 \cdot x^2 + 1 \cdot x^3 + 1 \cdot x^4 + \text{-----}$$

$$= x^3 + x^4 + x^5 + \text{----}$$

$$= x^3 [1 + x^2 + x^3 + \text{-----}]$$

$$= x^3 \cdot \frac{1}{(1-x)}$$

$$\boxed{G(x) = \frac{x^3}{(1-x)}}$$

2) $x = 0, 0, 1, 2, 3 \text{ -----}$

$$G(x) = 0 + 0x + 1 \cdot x^2 + 2 \cdot x^3 + 3 \cdot x^4 + \text{---}$$

$$= x^2 + 2x^3 + 3x^4 + \text{-----}$$

$$= x^2 [1 + 2x + 3x^2 + \text{-----}]$$

$$= x^2 \cdot \left[\frac{1}{(1-x)^2}\right]$$

$$\boxed{G(x) = \frac{x^2}{(1-x)^2}}$$

3) $x = 3, -3, 3, -3, 3, \text{ ------}$

$$G(x) = 3 - 3x + 3x^2 - 3x^3 + \text{-----}$$

$$= 3 [1 - x + x^2 - x^3 + \text{---}]$$

$$\boxed{G(x) = \frac{3}{1+x}}$$

4) $x = 3, 9, 27, 81 \text{------}$

$$G(x) = 3 + 9x + 27x^2 + 81x^3 + \text{------}$$

$$= 3(1 + 3x + 3^2x^2 + 3^3x^3 + \text{.....}]$$

$$= 3 \cdot \left(\frac{1}{1-3x}\right)$$

$$\boxed{G(x) = \frac{3}{1-3x}}$$

5) $x = 1, -2, 3, -4, \text{------}$

$$G(x) = 1 - 2x + 3x^2 - 4x^3 + 5x^4 - 6x^5 + \text{------}$$

$$= 1 + 2(-x) + 3(-x)^2 + 4(-x)^3 + 5(-x)^4 + \text{-----}$$

$$\boxed{G(x) = \frac{1}{(1+x)^2}}$$

Recurrence Relation :-

It is an Eq^n that recursively defines a sequence where the next term is a function of the previous terms.

Ex:- 1) $S = \{2, 2^2, 2^3, 2^4, \ldots\ldots 2^n \ldots\ldots\}$

$S = \{S_n\} = \{2^n\}$ → sequence

$\underbrace{S_n = 2S_{n-1} \quad , \quad n \geq 2}_{\text{Recurrence Relation}} \quad , \quad \underbrace{S_1 = 2}_{\text{Initial Condition}}$

∴ $\boxed{S_n = 2 \cdot S_{n-1} \text{ , } n \geq 2 \text{ is recurrence Relation with initial condition } S_1 = 2}$

2) $S = \{3, 7, 11, 15, 19, \ldots\ldots\ldots\}$

$S_n = S_{n-1} + 4 \quad , \quad n \geq 2 \quad , \quad S_1 = 3$

∴ $\boxed{S_n = S_{n-1} + 4 \text{ , } n \geq 2 \text{ is recurrence relation with initial condition , } S_1 = 3}$

3) $A = \{1, 1, 2, 3, 5, 8, 13, \ldots\ldots\}$

$\boxed{A_n = A_{n-1} + A_{n-2} \text{ , } n \geq 3 \text{ is recurrence relation with initial condition } A_1 = 1}$

A_n is known as fibonacci Sequence,

$A_n = A_{n-1} + A_{n-2} \quad , \quad n \geq 3 \quad , \quad A_1 = 1$

Degree :- Degree of linear recurrence relation is also 1.

It is defined as the highest power of a_n.

* Linear Recurrence Relation :-

A recurrence Relation having only the first power of the Sequence $\{a_n\}$ and not having any product term is said to be Linear.

The general form of linear recurrence relation with Constant Coefficient is given by;

$$C_0 a_r + C_1 a_{r-1} + C_2 a_{r-2} + C_3 a_{r-3} + \cdots + C_k a_{r-k} = f(r) \quad \text{--(1)}$$

where, $C_0, C_1, C_2 \cdots C_k$ are constants and $f(r)$ is a function of r (but not of a_r).

If both C_0 and C_k are non-zero then the recurrence relation in (1) is said to be of order "k"

Eqn (1) is called Linear Recurrence Relation with constant Coefficient.

Ex:-

$$2a_n + 3a_{n-1} = 3$$

$$a_n - 7a_{n-1} + 12a_{n-2} = n - 4^n$$

$$a_n = a_{n-1} + a_{n-2}$$

ORDER :- It is the diff b/w the height and lowest subscript of a Recurrence Relation

Recurrence Relation	order
$2a_n + 3a_{n-1} = 3$	$[n-(n-1)] = 1$
$a_n + 7a_{n-1} - 12a_{n-2} = 0$	$[n-(n-2)] = 2$
$a_n + a_{n-1} + a_{n-2} + a_{n-3} = 0$	$[n-(n-3)] = 3$
$a_n = 2a_{n-1} + 1$	1

Solution of Linear Recurrence Relation with constant Coefficient :-

There are multiple methods to solve linear recurrence Relation :-

* Iteration Method (not in course)
* Method of characteristic Roots
* generating functions

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} = f(n).$$

* Method of characteristic Roots.

Linear Recurrence Relation with const. coefficient

Homogeneous Recurrence Relation
↓
$f(n) = 0$

(Particular Solution) Non-Homogenous Recurrence Relation

$f(n) \neq 0$

[not included in syllybus all cases.]

① Homogenous Recurrence Relation :- $f(n) = 0$

*** steps to solve :-**

1. Put $a_n = r^n$, $a_{n-1} = r^{n-1}$ and so on
2. Find an Equation in terms of r. This is called as characteristic Equation or Auxillary Equation.
3. Solve characteristic Eq^n and Find characteristic Roots.
4. If characteristic Roots are

   * $r = r_1, r_2, r_3$ [Distinct]

general sol^n $\Rightarrow$ $a_n = b_1 r_1^n + b_2 r_2^n + b_3 r_3^n$,

where $\{ b_1, b_2, b_3$ are Constant

* $r = r_1, r_1$

$a_n = (b_1 + n b_2) r_1^n$

* $r = r_1, r_1, r_2, r_3$

$a_n = (b_1 + n b_2) r_1^n + b_3 r_2^n + b_4 r_3^n$

* $r = r_1, r_1, r_1, r_2$

$a_n = (b_1 + n b_2 + n^2 b_3) r_1^n + b_4 r_2^n$

Que:- Solve: $a_n = a_{n-1} + 2a_{n-2}$, $n \geq 2$ with the initial Condition; $a_0 = 0$, $a_1 = 1$.

Sol:- $a_n = a_{n-1} + 2a_{n-2}$

$a_n - a_{n-1} - 2a_{n-2} = 0$

put $a_n = r^n$, $a_{n-1} = r^{n-1}$, $a_{n-2} = r^{n-2}$

$r^n - r^{n-1} - 2 \cdot r^{n-2} = 0$

$r^n \left[1 - \frac{1}{r} - \frac{2}{r^2}\right] = 0$

$r^n \left[\frac{r^2 - r - 2}{r^2}\right] = 0$

$r^n [r^2 - r - 2] = 0$

$r^2 - r - 2 = 0$

$r = 2, -1$

when roots are Distinct then, $a_n = b_1 r_1^n + b_2 r_2^n$

$a_n = b_1 (2)^n + b_2 (-1)^n$

$\boxed{a_n = b_1 2^n + b_2 (-1)^n}$ — (1)

Now, put $a_0 = 0$

$$a_0 = b_1(2^0) + b_2(-1)^0$$

$$0 = b_1 \cdot 1 + b_2 \cdot 1$$

$$\boxed{0 = b_1 + b_2} \quad \text{— (ii)}$$

put $a_1 = 1$

$$a_1 = b_1(2^1) + b_2(-1)^1$$

$$1 = b_1(2) + b_2(-1)$$

$$\boxed{1 = 2 \cdot b_1 - b_2} \quad \text{— (iii)}$$

from Eq^n (ii) and (iii). By Elimination method.

$$\begin{aligned} b_1 + \cancel{b_2} &= 0 \\ 2b_1 - \cancel{b_2} &= 1 \\ \hline 3b_1 &= 1 \end{aligned}$$

$$\boxed{b_1 = \frac{1}{3}}$$

By Eq^n (ii) we get $\quad b_1 + b_2 = 0$

$$\frac{1}{3} + b_2 = 0$$

$$\boxed{b_2 = -\frac{1}{3}}$$

put these values in eq^n (i) we get.

$$a_n = \left(\frac{1}{3}\right)2^n + \left(-\frac{1}{3}\right)(-1)^n$$

$$\boxed{a_n = \frac{2^n}{3} - \frac{(-1)^n}{3}} \quad \text{— Ans}$$

Que :- Solve $a_n = 4(a_{n-1} - a_{n-2})$ with Initial Condition

$a_0 = a_1 = 1$

Sol :- $a_n = 4(a_{n-1} - a_{n-2})$

put $a_n = r^n$, $a_{n-1} = r^{n-1}$, $a_{n-2} = r^{n-2}$

$a_n - 4a_{n-1} + 4a_{n-2} = 0$

$r^n - 4 \cdot r^{n-1} + 4r^{n-2} = 0$

$r^n \left[1 - \frac{4}{r} + \frac{4}{r^2} \right] = 0$

$r^n \left[\frac{r^2 - 4r + 4}{r^2} \right] = 0$

$r^n [r^2 - 4r + 4] = 0$

$r^2 - 4r + 4 = 0$

$(r-2)^2 = 0$

$r = 2, 2$

when Roots are Similar, $a_n = (b_1 + n b_2) r_1^n$

$\boxed{a_n = (b_1 + n b_2) 2^n}$ — 1

put, $a_0 = 1$ and $a_1 = 1$ [initial Condition]

Now for $a_0 = 1$

$a_0 = (b_1 + 0 b_2) 2^0$

$1 = (b_1 + 0) \cdot 1$

$\boxed{1 = b_1}$

Now, put $a_1 = 1$

$a_1 = (b_1 + 1 \cdot b_2) 2^1$

$1 = (1 + b_2) 2$

$1 = 2 + 2b_2$

$1 - 2 = 2b_2$

$\boxed{-\frac{1}{2} = b_2}$

Now, put the values of b_1 and b_2 in Eq^n (1)

$$a_n = [b_1 + n b_2] 2^n$$

$$\boxed{a_n = \left[1 - \frac{n}{2}\right] 2^n} \text{ - Ans.}$$

Que :- Solve : $a_n - 8a_{n-1} + 21 a_{n-2} - 18 a_{n-3} = 0$

Sol :- put $a_n = r^n$, $a_{n-1} = r^{n-1}$, $a_{n-2} = r^{n-2}$,

$a_{n-3} = r^{n-3}$

$$a_n - 8 a_{n-1} + 21 a_{n-2} - 18 a_{n-3} = 0$$

$$r^n - 8 \cdot r^{n-1} + 21 \cdot r^{n-2} - 18 \cdot r^{n-3} = 0$$

$$r^n \left[1 - \frac{8}{r} + \frac{21}{r^2} - \frac{18}{r^3}\right] = 0$$

$$r^n \left[\frac{r^3 - 8r^2 + 21r - 18}{r^3}\right] = 0$$

$$r^n [r^3 - 8r^2 + 21r - 18] = 0$$

$$r^3 - 8r^2 + 21r - 18 = 0 \quad \text{— (1)}$$

Now, find the factors of Eq^n (1) we get

$$r = 3, 3, 2$$

when 2 roots are same and 1 is distinct then

$$a_n = (b_1 + n b_2) r_1^n + b_3 \cdot r_2^n$$

$$\boxed{a_n = (b_1 + n b_2) 3^n + b_3 \cdot 2^n} \text{ - Ans}$$

Ques Solve $a_n = -a_{n-1} + 4a_{n-2} + 4a_{n-3}$ with $a_0 = 8$, $a_1 = 6$, $a_2 = 26$

Sol $\rightarrow$

$$a_n + a_{n-1} - 4a_{n-2} - 4a_{n-3} = 0$$

$$\Rightarrow r^n + r^{n-1} - 4r^{n-2} - 4r^{n-3} = 0$$

$$\Rightarrow r^n\left[1 + \frac{1}{r} - \frac{4}{r^2} - \frac{4}{r^3}\right] = 0$$

$$\Rightarrow r^n\left[\frac{r^3 + r^2 - 4r - 4}{r^3}\right] = 0$$

$$\Rightarrow r^n\left[r^3 + r^2 - 4r - 4\right] = 0$$

$$r^3 + r^2 - 4r - 4 = 0$$

$$\Rightarrow r^2(r+1) - 4(r+1) = 0$$

$$\Rightarrow (r^2-4)(r+1) = 0$$

$$\Rightarrow r = -1, 2, -2.$$

$$\Rightarrow \boxed{a_n = b_1(-1)^n + b_2(2)^n + b_3(-2)^n} \longleftarrow (1)$$

put $a_0 = 8$

$a_0 = b_1(-1)^0 + b_2 2^0 + b_3(-2)^0$

$8 = b_1 + b_2 + b_3$ —— (ii)

put $a_1 = 6$

$a_1 = b_1(-1)^1 + b_2 2^1 + b_3(-2)^1$

$6 = -b_1 + 2b_2 - 2b_3$ —— (iii)

put $a_2 = 26$

$a_2 = b_1(-1)^2 + b_2 2^2 + b_3(-2)^2$

$26 = b_1 + 4b_2 + 4b_3$ —— (iv)

from eq (ii), (iii) and (iv)

$b_1 = 2$, $b_2 = 5$, $b_3 = 1$

put in eq (i)

$$\boxed{a_n = 2(-1)^n + 5(2)^n + (-2)^n}$$ Ans

Ques. for Practices

1) Solve $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with $a_0 = 1$, $a_1 = -2$, $a_2 = -1$

Ans → $a_n = (1 + 3n - 2n^2)(-1)^n$ Ans

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2) Solve $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1$, $a_1 = 6$

Ans → $a_n = (1+n)3^n$ Ans

* Particular Solution of Reccurence Relation.

Case I :- $f(n) = p(n) \mid f(n) \neq 0$

where, $p(n)$ is a polynomial of degree S.

Steps to find $a_n^{(P)}$:-

1) Let $a_n = A_0 + A_1 n + A_2 n^2 + \cdots\cdots A_s n^s$.
2) Put value of $a_n, a_{n-1}, a_{n-2} \cdots\cdots$ so on in the given Eq^n
3) Compare the coefficient of like powers on n.
4) Find $A_0, A_1, A_2 \cdots\cdots A_s$.
5) $a_n^{(P)} = A_0 + A_1 n + A_2 n^2 + A_3 n^3 + \cdots\cdots A_s n^s$

{By putting the values of A's}

Que :- Solve :- $y_{n+2} - y_{n+1} - 2y_n = n^2$

NOTE :- GENERAL SOLUTION | TOTAL Solution

$$a_n = a_n^{(h)} + a_n^{(P)}$$

where, $a_n^{(h)}$ = homogeneous Solution

$a_n^{(P)}$ = Particular Solution

Sol :- To find $y_n^{(h)}$: $y_{n+2} - y_{n+1} - 2y_n = 0$

{order $= (n+2) - n = 2$}

Now, Put $y_n = r^n$

characteristic Eq^n $= r^2 - r - 2 = 0$

$r = -1, 2$

$$\boxed{y_n^{(h)} = b_1 (-1)^n + b_2 (2)^n}$$

To find $y_n^{(P)}$:- $y_{n+2} - y_{n+1} - 2y_n = n^2$

put $y_n = A_0 + A_1 n + A_2 n^2$

{2 degree polynomial in RHS}

we have, $y_{n+2} - y_{n+1} - 2y_n = n^2$

$$[A_0 + A_1(n+2) + A_2(n+2)^2] - \{A_0 + A_1(n+1) + A_2(n+1)^2\} -$$
$$2\{A_0 + A_1 n + A_2 n^2\} = n^2$$

$$\{A_0 + A_1(n+2) + A_2(n^2+4+4n)\} - \{A_0 + A_1(n+1) + A_2(n^2+1+2n)\}$$
$$- 2\{A_0 + A_1 n + A_2 n^2\} = n^2$$

Simplify;

$$\{A_1 - 2A_0 + 3A_2\} + \{2A_2 - 2A_1\}n + \{-2A_2\}n^2 = n^2$$

comparing coefficient

(const) $A_1 - 2A_0 + 3A_2 = 0$ — (1)

(deg.1) $2A_2 - 2A_1 = 0$ — (2)

(deg.2) $-2A_2 = 1$ — (3)

$$\boxed{A_2 = -\frac{1}{2}}$$

$$\boxed{A_1 = -\frac{1}{2}}$$ from eq^n. (2)

$$\boxed{A_0 = -1}$$ from eq^n (1)

Now, for particular Solution,

$$y_n^{(P)} = A_0 + A_1 n + A_2 n^2$$

$$\boxed{y_n^{(P)} = -1 - \frac{n}{2} - \frac{n^2}{2}}$$ — Ans

Now, general solution, $y_n = y_n^{(h)} + y_n^{(P)}$

$$\boxed{y_n = b_1(-1)^n + b_2(2)^n - 1 - \frac{n}{2} - \frac{n^2}{2}}$$ — Ans

Que :- Solve $a_{n+2} + 2a_{n+1} - 15a_n = 6n + 10$ with initial Condition $a_0 = 1$, $a_1 = -\frac{1}{2}$.

Sol :-

To find $a_n^{(h)}$: $a_{n+2} + 2a_{n+1} - 15a_n = 0$

{order $= (n+2) - n = 2$}

Now put $a_n^{(h)} = r^n$

characteristic Eq^n $\Rightarrow$ $r^2 + 2rn - 15 = 0$

$r = -5, 3$

$$\boxed{a_n^{(h)} = b_1(-5)^n + b_2(3)^n}$$

To find $a_n^{(P)}$:

$$a_{n+2} + 2a_{n+1} - 15a_n = 6n + 10$$

put $a_n = A_0 + A_1 n$

{polynomial of degree 1 in RHS}

$$\{A_0 + A_1(n+2)\} + 2\{A_0 + A_1(n+1)\} - 15\{A_0 + A_1 n\} = 6n + 10$$

Simplify :

$$\{-12A_0 + 4A_1\} + ((-12)A_1 n = 6n + 10$$

Comparing Coefficient.

Const. $\quad -12A_0 + 4A_1 = 10$ — (I)

(deg 1) $\quad (-12) A_1 = 6$ — (II)

$$\boxed{A_1 = -\frac{1}{2}}$$

$$\boxed{A_0 = -1}$$ from Eq^n (I)

$$a_n^{(P)} = A_0 + A_1 n$$

$$\boxed{a_n^{(P)} = -1 - \frac{n}{2}}$$

{$\therefore$ gen. Sol^n = $a_n = a_n^{(h)} + a_n^{(P)}$}

General Sol^n :- $\boxed{a_n = b_1(-5)^n + b_2(3)^n - 1 - \frac{n}{2}}$ — (III)

Now we have initial Conditions also ∴ So we put the values in Eq^n (III)

$a_0 = 1$

$1 = b_1(-5)^0 + b_2(3)^0 - 1 - \frac{0}{2}$

$\boxed{b_1 + b_2 = 2}$

$a_1 = -\frac{1}{2}$

$-\frac{1}{2} = (-5)^1 b_1 + b_2(3)^1 - 1 - \frac{1}{2}$

$\boxed{-5b_1 + 3b_2 = 1}$

By Elimination Method,

$b_1 = \frac{5}{8}$, $b_2 = \frac{11}{8}$

Put in general Sol^n :-

$\boxed{a_n = \frac{5}{8}(-5)^n + \frac{11}{8}(3)^n - 1 - \frac{n}{2}}$ — Ans

*

Solution of Reccurence Relation by Generating Function

STEPS :-

1) In given Equation multiply by x^r (If variable is r).

2) Take summation from $r=1$ to ∞, if 1 initial condition is given.

3) Take summation from $r=2$ to ∞, if 2 initial Cond. is given and so on.

4) write each summation in terms of $G(x)$ or closed form

5) Put values of each summation and find $G(x)$.

6) write a_r :-

$$\boxed{* \; G(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots\cdots}$$

Que :- Solve: $a_r - 2a_{r-1} - 3a_{r-2} = 0$, $a_0 = 3$, $a_1 = 1$ by using generating function.

Sol :- $a_r - 2a_{r-1} - 3a_{r-2} = 0$

multiply Eq^n By x^r we get

$$a_r \cdot x^r - 2a_{r-1} \cdot x^r - 3a_{r-2} \cdot x^r = 0$$

Now, we have 2 initial condition then, we take Summation from $r = 2$ to ∞.

$$\sum_{r=2}^{\infty} a_r \cdot x^r - 2\sum_{r=2}^{\infty} a_{r-1} x^r - 3\sum_{r-2}^{\infty} a_{r-2} \cdot x^r = 0$$

Take, $\displaystyle\sum_{r=2}^{\infty} a_r x^r = a_2 \cdot x^2 + a_3 x^3 + \cdots\cdots$

$$\left\{ \begin{array}{l} G(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \cdots\cdots \\ G(x) - a_0 - a_1 x = a_2 x^2 + a_3 x^3 + \cdots\cdots \end{array} \right\}$$

$$= G(x) - a_0 - a_1 x \qquad \because a_0 = 3, a_1 = 1$$

$$\boxed{\sum_{r=2}^{\infty} a_r \cdot x^r = G(x) - 3 - x}$$

Now we take,

$$\sum_{r=2}^{\infty} a_{r-1} \cdot x^r = a_1 x^2 + a_2 x^3 + a_3 x^4 + \text{------}$$

$$= x\left[a_1 x + a_2 x^2 + a_3 x^3 + \text{----}\right]$$

$$= x \cdot \left[G(x) - a_0\right]$$

$$\boxed{\sum_{r=2}^{\infty} a_{r-1} \cdot x^r = x \cdot G(x) - 3}$$

Now we take,

$$\sum_{r=2}^{\infty} a_{r-2} \cdot x^r = a_0 x^2 + a_1 x^3 + a_2 x^4 + \text{-----}$$

$$= x^2 \left[a_0 + a_1 x + a_2 x^2 + \text{------}\right]$$

$$= x^2 \, G(x)$$

Now put all the values in eq^n (1)

$$G(x) - a_0 - a_1 x - 2x\left[G(x) - a_0\right] - 3x^2 (G(x) = 0$$

$$G(x) - 3 - x - 2x\,G(x) - 2x \cdot 3 - 3x^2 G(x) = 0$$

$$G(x)\left[1 - 2x - 3x^2\right] = 3 + x - 6x$$

$$G(x)\left[1 - 2x - 3x^2\right] = 3 - 5x$$

$$G(x) = \frac{3 - 5x}{(1 - 2x - 3x^2)}$$

Now, we have to solve it By partial fraction;

$$\left\{\begin{array}{l} 1 - 2x - 3x^2 \\ -(3x^2 + 2x - 1) \\ 3x(x+1) - 1(x+1) \\ -(x+1)(3x-1) \end{array}\right.$$

$$G(x) = \frac{5x - 3}{(x+1)(3x-1)}$$

$$G(x) = \frac{2}{x+1} - \frac{1}{3-1}$$

formula

$$\boxed{a_r = 2 \cdot (-1)^r - 3^r} \text{ — Ans}$$

H.W

Que 2 :- Solve $a_r - a_{r-1} - 2a_{r-2} = 0$ with initial cond. $a_0 = 0$, $a_1 = 1$ By generating function Solⁿ.

Que 3 :- Solve: $a_{n+2} + a_n = 2^n$ with initial cond. $a_0 = 2$, $a_1 = 1$ By generating functⁿ Solⁿ.

$\times$

Course Complete.

$\times$